Stephanie Buttigieg

PhD Student · Institute of Astronomy, University of Cambridge

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Research interests

- The formation and evolution of supermassive black holes
- Cosmological simulations of galaxy formation and evolution
- Observations of merging black holes using gravitational waves

Education

Institute of Astronomy, University of Cambridge — *Cambridge, UK*
PhD in Astronomy — *Oct 2023–Aug 2027*

- Supervisors: Prof. Debora Sijacki and Dr Christopher J. Moore
- Research area: Supermassive black hole mergers in cosmological simulations
- Core member, LISA Consortium

University of Cambridge — *Cambridge, UK*
MASt in Astrophysics — *Oct 2022–Jun 2023*

- Graduated with distinction
- Final project: *Is space haunted? Exorcising ghosts from the gravitational particle spectrum*

University of Malta — *Msida, Malta*
BSc in Mathematics and Physics — *Oct 2018–Sep 2022*

- Graduated with first class honours; ranked top of class
- Exchange year at the University of Warwick, UK (Sep 2020–Jun 2021)
- Final year project: *Dynamical Systems in Beyond Standard Theory Cosmology*

Publications

Published

Buttigieg, S., Sijacki, D., Moore, C. J., Bourne, M. A., 2025. *Premature supermassive black hole mergers in cosmological simulations of structure formation*. Monthly Notices of the Royal Astronomical Society, 542(3), 2019–2038. doi:10.1093/mnras/staf1336. e-Print: arXiv:2504.17549.

Buttigieg, S., Sijacki, D., Moore, C. J., Bourne, M. A., Sesana, A., 2026. *Comparing gravitational wave background predictions from cosmological simulations to pulsar timing observations*. Monthly Notices of the Royal Astronomical Society, 552(3), stag1749. doi:10.1093/mnras/stag1749. e-Print: arXiv:2607.05208.

Cooke, G. J.*, White, S. B.*, **Buttigieg, S.***, Lowson, N., Galaudage, S., Battisti, A., Kelsey, L. & Ross, K., 2026. *The absence of women's scientific contributions in core UK pre-university science qualifications.* Published in the British Educational Research Journal, 00, 1–22. doi:10.1002/berj.70279. *Joint first authors.[1]

Pre-Prints

Buttigieg, S., Madigan, M., Kamalaruban P., Burrel S., 2026. *Latent space bias directions in LLMs capture confidence, not fairness*. To be submitted to the ACM Conference on Fairness, Accountability, and Transparency. e-Print:arXiv:2610.08559

Presentations

Invited talks

Sep 2026	Kavli Astrophysics Symposium 2026, Cambridge, UK
Jul 2026	Computational Galaxy Formation Workshop, Germany
May 2026	University of Helsinki Astrophysics Seminar
Jan 2026	STScI Galaxy & AGN Journal Club (teleconference)
Nov 2025	Galaxy and SMBH formation, growth, and co-evolution workshop, Nepal
May 2025	NANOGrav Astrophysics Working Group (teleconference)
Mar 2025	CamGW Meeting, Kavli Institute for Cosmology, University of Cambridge

Contributed talks

Oct 2026	Young astronomers on galactic nuclei meeting, Seville, Spain

[1] This publication received media coverage, including from the BBC.

Mar 2026 Emerging Research Seminar Series, Christ's College, University of Cambridge
Mar 2026 CamGW Meeting, Kavli Institute for Cosmology, University of Cambridge
Sep 2025 Black holes across cosmic time conference, Cambridge, UK
Sep 2025 Young astronomers on galactic nuclei meeting, Southampton, UK
Jul 2025 GR–Amaldi conference, Glasgow, UK
May 2025 Unveiling massive black hole evolution with gravitational waves and light workshop, Paris, France
Feb 2025 Wednesday Seminar Series, Institute of Astronomy, University of Cambridge

Awards, Fellowships, & Grants

2023–27 **Cambridge Centre for Doctoral Training in Data-Intensive Science PhD Studentship**, UK STFC ~£20,000 p.a.
2022–23 **Tertiary Education Scholarship Scheme**, Government of Malta £10,000
2025 **Education and Outreach Grant**, Royal Astronomical Society, IncludeHer UK collaboration £5,000
2023 **Field's Scholarship**, Christ's College Cambridge
2026 **Community Service Award**, Institute of Astronomy, University of Cambridge
2023 **Evolve Prize for Science**, University of Malta
2019, 20, 22 **Dean's List**, Faculty of Science, University of Malta
2026 **Monica Kornberg Fund**, Christ's College, Cambridge
2025, 2026 **Bottomley Fund for Travel**, Christ's College, Cambridge

Teaching & Admissions Experience

2025–26 **Mathematics for Natural Science Tripos Part Ia**, Supervisor, Christ's College, University of Cambridge
2023–25 **Applied Mathematics**, Undergraduate admissions interviewer, Christ's College, University of Cambridge
2018–24 **Mathematics and Physics**, Private Tutor
2019 **PwC Malta**, STEM Trainer

Professional & Industry Experience

Visa
Research Intern *Mar 2026–Sep 2026*
LLM interpretability and debiasing via activation steering
European Organization for Nuclear Research (CERN)
Summer Research Intern *Jun 2022–Aug 2022*
Data quality tests for the FLUKA software

Skills

Programming
Python (NumPy, SciPy, Astropy, PyTorch, Matplotlib), Mathematica, parallel coding, Bash
Computing
HPC cluster computing, GPU computing, Git
Cosmological simulations
Post-processing and analysis of simulation outputs
Statistical & machine learning
Bayesian inference and MCMC, large language model interpretability
Languages
Maltese and English (native), Italian (basic conversational)

Service, Outreach & Leadership

Science Communication & Broadening Participation
Ongoing Creator and presenter of *Astronomija bil-Malti*, an online science communication series in Maltese, with over 350,000 views in total.
Jan 2026 Public speaker, Institute of Astronomy Open Evenings. Talk: *Listening to the Universe with Gravitational Waves*.
Mar 2025 Invited panellist, *Her Voice in Science*, European Parliament.

Feb 2025 Invited panellist, *Curiosity Knows No Boundaries: Celebrating Women and Girls in Science*, Malta Chamber of Scientists, University of Malta.

Oct 2024 Public speaker, Institute of Astronomy Open Evenings. Talk: *Meet the Cosmic Giants: Understanding Supermassive Black Holes*.

Professional Appointments

Mar 2026–Sep 2026 Local Organising Committee member, *Kavli Astrophysics Symposium 2026*.

Sep 2025–Sep 2026 Local Organising Committee member, *Black holes across cosmic time*.

Jan 2025–Jan 2026 Co-chair, Women's Network, Institute of Astronomy.

Oct 2024–Oct 2025 PhD Academic Representative, Faculty Board for Physics and Chemistry, University of Cambridge.

Oct 2022–Mar 2024 International Officer; Women's & Non-Binary Officer; Vice President, Christ's College MCR.

Oct 2021–Jun 2022 Student Representative, Board of Studies, Department of Physics, University of Malta.

Jul 2019–Jun 2020 Education Officer, Science Students' Society, University of Malta.